Applied Energy – Submission Highlights

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Project REGENESIS: Autonomous Multi-Physics Self-Regulation and Thermal-Stress Bounding in High-Temperature Solid-Oxide CO_2 Electrolysis and Sabatier Synthesis Loops

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Core Highlights (Max 85 characters per bullet point including spaces):

- Formulated unified multi-physics Master Jacobian framework for solid-oxide SOEC loops. (84 chars)
- Proved autonomous thermochemical self-regulation via Re(lambda_max) = -0.1389 ¡ 0. (81 chars)
- Resolved microsecond kinetic stiffness via Strang splitting (CVODE + CFL r ¡= 0.50). (83 chars)
- Confirmed 100% stability across 5 operational folds in LOCO-CV (RMSE ¡= 3.47%). (79 chars)
- Bounded 10,000-hr accelerated degradation loss at ¡1.85% per 1,000 operating hours. (82 chars)